

# NORMAL-FORM COMPLEXITY OF A THREE-SHEETED KELLER COVER

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ABSTRACT. Let  $K = X + Q + C$  be a degree-at-most-three Keller map in dimension  $n$ , with  $Q, C$  homogeneous of degrees two and three. If the coordinate span of  $C$  has dimension  $r$ , we give a collision-preserving cubic-homogeneous suspension in dimension  $n + r + 1$ . Applied to the public 11-variable degree-three descendant of the new Jacobian counterexample, where  $r = 7$ , this recovers the 19-variable construction independently announced by Harris Chan.

We determine the generic Jordan type of its nonlinear Jacobian as  $(18, 1)$  and give a two-point certificate that its nilpotent flag moves. Symmetric doubling gives an explicit Hessian-nilpotent quartic in 38 variables, with generic Hessian type  $(35, 2, 1)$ . A two-parameter family of algebraic inverse rays shows

$$\Delta^m(Q^{m+1}) \neq 0 \quad \text{for every } m \geq 1,$$

and identifies the inverse singularities with intersections with the non-properness hypersurface. For the fixed 19-variable cubic tensor, we construct a 110-variable square-zero cubic-linear pairing and prove that every vector-Waring decomposition uses at least 52 cubes. Finally, an exact 13-term functional is a constant nonzero quartic obstruction across a natural 20-dimensional equivariant family of rank-six cubic compressions. The example-specific statements are computer-assisted and come with exact rational certificates.

*Working computer-assisted manuscript. The recovered exact certificates have been rerun. Independent CAS reproduction and specialist review remain submission requirements.*

## 1. INTRODUCTION

The classical reductions of Bass–Connell–Wright and Yagzhev make cubic-homogeneous maps central to the Jacobian conjecture [3]. Once a three-dimensional counterexample was announced [1], it therefore became natural to ask how economically that counterexample reaches the classical normal forms. We credit Akhil Mathew as the primary human source of the problem: Mathew prompted Levent Alpöge, Alpöge prompted Fable, Fable produced the example, and Alpöge announced it publicly [9].

Atkins-Turkish publicly supplied a degree-at-most-three map in 11 variables, with three explicit colliding points [2]. Its cubic coordinate space

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*Date:* July 22, 2026.

has dimension seven. The construction in section 2 consequently gives a cubic-homogeneous counterexample in

$$11 + 7 + 1 = 19$$

variables. Harris Chan independently announced this special sparse lift on July 23, 2026 [4]; we make no sole-priority claim for the 19-variable endpoint. The reusable contribution is the rank-sensitive  $n + r + 1$  theorem.

The fixed 19-variable tensor has further structure. Its nonlinear Jacobian has generic Jordan type  $(18, 1)$ . Classical symmetric reduction [5] then produces a quartic potential in 38 variables. A 48-variable explicit Hessian witness was already public through the work of Santib   ez-Leal, starting from Thompson’s 24-variable cubic map [8, 7]. Our claims are the quantitative dimension 38, the exact Hessian Jordan type  $(35, 2, 1)$ , and the all-order Laplacian formula.

In the cubic-linear direction, we use the pairing formalism of Gorni and Zampieri [6]. The existence of such pairings and the square-zero mechanism are not new. For this particular tensor we prove the model-relative interval

$$52 \leq N_{\text{pair}} \leq 110.$$

Neither endpoint is asserted to be sharp for arbitrary cubic-homogeneous or cubic-linear counterexamples.

The main results are summarized as follows.

**Theorem 1.1** (Rank-sensitive suspension). *Let  $K = X + Q + C: \mathbb{A}^n \rightarrow \mathbb{A}^n$  be Keller over a characteristic-zero field, with  $Q$  quadratic homogeneous and  $C$  cubic homogeneous. Put*

$$r = \dim \text{span} \{C_1, \dots, C_n\}.$$

*Then  $K$  admits a cubic-homogeneous Keller suspension in dimension  $n+r+1$ . Every collision of  $K$  lifts explicitly to a collision of the suspension.*

**Theorem 1.2** (The fixed 19-variable tensor). *For the tensor  $H$  of section 3:*

- (i)  *$I + H$  is a noninjective cubic-homogeneous Keller map in 19 variables;*
- (ii) *the generic Jordan type of  $JH$  is  $(18, 1)$ ;*
- (iii) *the associated 38-variable quartic  $\mathcal{Q}$  is Hessian nilpotent, has generic Hessian type  $(35, 2, 1)$ , and violates Zhao’s eventual vanishing conclusion at every order; and*
- (iv) *the vector-Waring length of  $H$  is at least 52, while  $H$  has a full-rank square-zero pairing of length 110.*

**Theorem 1.3** (Equivariant compression obstruction). *Let  $V$  be the space of quadratic source fields that preserve the weights; it has dimension 115. The locus in  $V$  killing five specified cubic output rows contains a natural affine slice of dimension 20. Each point on that slice has at most six linearly independent cubic coordinate functions. The same functional, supported on 13 terms, takes the value 1 on its quartic obstruction. Thus no point lifts*

through quartic order in the stated equivariant conjugacy model. The original 605-shift statement is a sparse corollary.

Theorem 1.3 is deliberately model-relative. It does not prove that 19 variables are globally minimal.

## 2. A RANK-SENSITIVE CUBIC SUSPENSION

Let  $k$  be a field of characteristic zero and write

$$K(X) = X + Q(X) + C(X),$$

where  $Q$  and  $C$  are homogeneous of degrees two and three. Choose a basis  $q = (q_1, \dots, q_r)^T$  of the coordinate span of  $C$ , and a constant matrix  $B \in M_{n \times r}(k)$  such that

$$C = Bq.$$

Introduce  $w = (w_1, \dots, w_r)$  and define

$$\widetilde{K}(X, w) = (X + Q(X) + Bw, w - q(X)). \quad (2.1)$$

**Lemma 2.1.** *The map  $\widetilde{K}$  is polynomially stably left-right equivalent to  $K$ .*

*Proof.* Set

$$S(X, w) = (X, w - q(X)), \quad T(U, V) = (U + BV, V).$$

Both are triangular polynomial automorphisms, and direct substitution gives

$$\widetilde{K} = T \circ (K \times \text{id}_{\mathbb{A}^r}) \circ S.$$

□

Adjoin a homogenizing coordinate  $t$ , and put

$$G(X, w, t) = (X + tQ(X) + t^2Bw, w - q(X), t). \quad (2.2)$$

Its nonlinear part is cubic homogeneous.

*Proof of theorem 1.1.* The Jacobian matrix in the  $X, w$  coordinates has block form

$$\begin{pmatrix} I + tJQ(X) & t^2B \\ -Jq(X) & I \end{pmatrix}.$$

The Schur complement gives

$$\begin{aligned} \det JG &= \det(I + tJQ(X) + t^2BJq(X)) \\ &= \det JK(tX) = 1. \end{aligned} \quad (2.3)$$

If  $K(p) = K(p')$ , then

$$\widehat{p} = (p, q(p), 1), \quad \widehat{p}' = (p', q(p'), 1)$$

are distinct and satisfy

$$G(\widehat{p}) = (K(p), 0, 1) = G(\widehat{p}').$$

Finally, write  $G = I + H$ . Since  $H$  is cubic homogeneous,

$$JH(sZ) = s^2JH(Z).$$

The identity  $\det(I + JH) = 1$  therefore implies  $\det(I + \lambda JH) = 1$  as a polynomial identity in  $\lambda$ . All nonconstant characteristic coefficients vanish, so  $JH$  is nilpotent.  $\square$

*Remark 2.2.* For a fixed  $K$ , any factorization  $C = B'q'$  through  $s$  scalar cubic coordinates has  $s \geq r$ . Thus  $r$  auxiliaries are optimal within the factorized suspension (2.1). This says nothing about other stable representatives of  $K$  or other homogenization models.

### 3. THE FIXED 19-VARIABLE TENSOR

We now work over  $\mathbb{Q}$ . In the coordinate order

$$X = (x, y, z, a, b, c, d, q, s, h, k),$$

let  $\Phi = (\Phi_1, \dots, \Phi_{11})$  be the public map [2]:

$$\begin{aligned} \Phi_1 &= -ac - adz - 3ay^2 - 2az - cd^2 + d^2z - ds + 7dy^2 \\ &\quad + sxy + 3xyz + 4y^2 + z, \\ \Phi_2 &= -bc - bdz - 3by^2 - 2bz - 3cdx - dq + qxy \\ &\quad + 12xy^2 + 3xz + y, \\ \Phi_3 &= -hk - hxz + kx^2 - 3x^2y + 2x, \\ \Phi_4 &= a - d^2 + 2dxy, \quad \Phi_5 = b + 3x^2y, \\ \Phi_6 &= c + xyz + 3y^2 + 2z, \quad \Phi_7 = d - xy, \\ \Phi_8 &= bz + 3cx + q, \\ \Phi_9 &= s + az + cxy - xyz - 7y^2 + cd - dz, \\ \Phi_{10} &= h - x^2, \quad \Phi_{11} = k + xz. \end{aligned}$$

Its linear part  $L = J\Phi(0)$  has determinant  $-2$ . Normalize by

$$K = L^{-1}\Phi = X + Q + C.$$

The coordinate cubics

$$q_1 = C_1, \quad q_2 = C_2, \quad q_3 = C_3, \quad q_4 = C_4, \quad q_5 = C_5, \quad q_6 = C_6, \quad q_7 = C_9$$

are linearly independent, and the remaining coordinate cubics vanish. Thus  $C = Bq$ , where  $B$  includes these seven coordinates into  $\mathbb{Q}^{11}$ .

With  $w = (w_1, \dots, w_7)$ , define

$$H(X, w, t) = (tQ(X) + t^2Bw, -q(X), 0) \tag{3.1}$$

in the coordinate order

$$Z = (x, y, z, a, b, c, d, q, s, h, k, w_1, \dots, w_7, t).$$

Then  $G = I + H$  is the suspension (2.2).

For reference, three colliding points of  $\Phi$  are

$$\begin{aligned} p_1 &= (0, 0, -1/4, 0, 0, 1/2, 0, 0, 0, 0), \\ p_2 &= (1, -3/2, 13/2, -9/4, 9/2, -10, -3/2, 3/4, -153/8, 1, -13/2), \\ p_3 &= (-1, 3/2, 13/2, -9/4, -9/2, -10, -3/2, -3/4, -153/8, 1, 13/2). \end{aligned}$$

They all map to  $(-1/4, 0, \dots, 0)$ . Consequently

$$(p_j, q(p_j), 1), \quad j = 1, 2, 3,$$

are three colliding points of  $G$ .

**3.1. Generic Jordan type.** Let  $A = JH$ , and set

$$E = xy(c + 2z) + d^2z + 3dy^2.$$

Define two polynomial vectors

$$v_1 = 2tx e_h + 2tz e_k + (xk + zh)e_{w_1}, \quad (3.2)$$

$$v_2 = -txy e_a - t(dz + 3y^2)e_s - Ee_{w_3} + 2Ee_{w_6} + xyz e_{w_7}. \quad (3.3)$$

**Proposition 3.1.** *Over  $\mathbb{Q}(Z)$ , the matrix  $A$  has Jordan type  $(18, 1)$ .*

*Proof.* Exact differentiation gives

$$Av_1 = Av_2 = 0.$$

The vectors are generically independent, while the all-ones specialization of  $A$  has rank 17. Hence the generic kernel has dimension two.

Let  $e_t$  be the last coordinate vector. Exact iteration gives

$$A^{17}e_t = 18t^{15}x^9y^5z^2v_2 \neq 0. \quad (3.4)$$

By theorem 1.1,  $A$  is nilpotent. A nilpotent 19-by-19 matrix with a two-dimensional kernel has two Jordan blocks, and (3.4) forces the larger block to have size at least 18. Thus the partition is  $(18, 1)$ .  $\square$

*Remark 3.2.* At the all-ones specialization the ranks of  $A, A^2, \dots, A^{18}$  are

$$17, 16, \dots, 1, 0.$$

The vectors

$$e_t, Ae_t, \dots, A^{17}e_t, v_1$$

specialize to a basis with determinant 1952152956156672.

#### 4. A RESTRICTED COMPRESSION OBSTRUCTION

Consider quadratic near-identity source conjugacies. At cubic order their action is

$$C \mapsto C + [Q, P_2], \quad [Q, P_2] = JQ P_2 - J P_2 Q.$$

The shift

$$P_2 = -d^2e_a$$

lowers the cubic coordinate-span rank from seven to six. Its exact conjugate has quartic remainder

$$(D_4)_z = d^3z + 3d^2y^2, \quad (D_4)_a = x^2y^2, \quad (D_4)_c = -2d^3z - 6d^2y^2. \quad (4.1)$$

For a quartic vector field  $R$ , define

$$\begin{aligned} \Lambda_4(R) = & [x^2y^2]R_a + 4[x^2ac]R_a - \frac{20}{3}[x^2as]R_a + 9[x^2qk]R_a \\ & + 3[xchk]R_a - \frac{1}{2}[xd^2q]R_a - 7[xshk]R_a + \frac{4}{3}[y^2ah]R_a \\ & - \frac{1}{2}[ybch]R_a - [ydhk]R_a + \frac{8}{3}[za^2h]R_a + 25[aqhk]R_a - 2[dqhk]R_d. \end{aligned} \quad (4.2)$$

**Proposition 4.1.** *For every cubic vector field  $P_3$ ,*

$$\Lambda_4([Q, P_3]) = 0, \quad \Lambda_4(D_4) = 1.$$

*Thus  $D_4 \notin \text{im}[Q, -]$ .*

*Moreover, among the 605 shifts*

$$P_2 = \lambda m e_j,$$

*where  $m$  is a quadratic monomial not involving  $X_j$ , the shift  $-d^2e_a$  is the unique one that lowers the cubic span to six.*

*Proof.* The first statement is finite coefficient linear algebra. Evaluating (4.2) on all

$$11 \binom{13}{3} = 3146$$

cubic monomial vector fields gives zero, while direct evaluation on (4.1) gives one. The restricted homological matrix on the 13 displayed coefficient positions has rank 12 and augmented rank 13.

For the second statement, a fixed nonzero seven-by-seven minor of the cubic coefficient matrix is evaluated along each of the 605 directions. Of these, 583 have no exceptional parameter and 22 give 27 rational roots. Exact full-rank evaluation at every root leaves only  $P_2 = -d^2e_a$ .  $\square$

**Proposition 4.2.** *Use the quadratic output corrections generated by the four outputs  $d, q, h, k$ , which have degree at most two after the conjugacy. Their ten quadratic products create a ten-dimensional quartic image. The  $z$ - and  $c$ -components of  $D_4$  lie outside this image. The  $a$ -component has the unique preimage  $-Y_d^2$ , but applying it restores cubic coordinate-span rank seven.*

*Proof.* This is another exact coefficient-matrix calculation. The ten quartic images are independent. Solving against the three nonzero components in (4.1) gives the assertions.  $\square$

## 5. THE 38-VARIABLE HESSIAN QUARTIC

Let  $F(x) = x + H(x)$  be the 19-variable map. On  $\mathbb{A}_x^{19} \times \mathbb{A}_y^{19}$ , define the cotangent double

$$\Phi_F(x, y) = (F(x), JF(x)^T y).$$

Its Jacobian is block lower triangular, with diagonal blocks  $JF$  and  $JF^T$ , so  $\det J\Phi_F = 1$ .

Over  $\mathbb{Q}(i)$ , put

$$T(u, v) = \left( u + iv, \frac{u - iv}{2} \right).$$

The scalar block satisfies

$$TT^T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Therefore

$$\mathcal{G} = T^{-1}\Phi_F T = I + \nabla \mathcal{Q},$$

where

$$\mathcal{Q}(u, v) = \frac{1}{2}(u - iv)^T H(u + iv). \quad (5.1)$$

This is a homogeneous quartic with 724 nonzero monomials. If  $F(p) = F(p')$ , then

$$\Phi_F(p, 0) = \Phi_F(p', 0),$$

so  $\mathcal{G}$  has an explicit transported collision.

**Proposition 5.1.** *The generic Jordan type of  $\text{Hess } \mathcal{Q}$  is*

$$(35, 2, 1).$$

*Proof.* Before the linear twist, the nonlinear Jacobian is

$$N = \begin{pmatrix} A & 0 \\ \mathcal{L} & A^T \end{pmatrix}, \quad A = JH(x), \quad \mathcal{L} = \text{Hess}_x(y^T H(x)).$$

The Hessian of  $\mathcal{Q}$  is similar to  $N$ .

The kernel vectors (3.2)–(3.3) satisfy

$$D^2 H[v_1, v_2] = D^2 H[v_2, v_2] = 0.$$

Since they span  $\ker A$  generically, this implies  $\mathcal{L}v_2 \in \text{im } A^T$ . In addition to the two kernel vectors  $(0, \ker A^T)$ , the matrix  $N$  therefore has a kernel vector whose first component is  $v_2$ . Hence  $\text{rank } N \leq 35$ . An exact all-ones specialization has rank 35.

For  $k \geq 1$ , the lower-left block of  $N^k$  is

$$\sum_{j=0}^{k-1} (A^T)^j \mathcal{L} A^{k-1-j}.$$

Since  $A^{18} = 0$ , the only possible term for  $k = 35$  is  $(A^T)^{17} \mathcal{L} A^{17}$ . The image of  $A^{17}$  is spanned by  $v_2$ , and  $D^2 H[v_2, v_2] = 0$ , so this term vanishes. Thus  $N^{35} = 0$ . At the all-ones specialization,

$$(N^{34})_{38,19} = 648 \neq 0.$$

The nilpotency index is 35 and the rank is 35, so there are three blocks. The remaining three dimensions split as  $2 + 1$ , proving the result.  $\square$

**5.1. An all-order Laplacian certificate.** Let

$$\bar{Y} = (1/2, 1, 0, \dots, 0, 1) \in \mathbb{Q}^{19}, \quad \bar{W} = T^{-1}(\bar{Y}, 0) = \left( \frac{\bar{Y}}{2}, -\frac{i\bar{Y}}{2} \right).$$

An exact inverse branch of the original three-dimensional map, lifted through the stable reductions, gives

$$(F^{-1}(s\bar{Y}))_1 = \frac{1/\sqrt{1-s^4}-1}{s^3} = \sum_{r \geq 1} \frac{1}{4^r} \binom{2r}{r} s^{4r-3}. \quad (5.2)$$

Consequently

$$(\mathcal{G}^{-1}(s\bar{W}))_{u_1} = \frac{1}{2} \sum_{r \geq 1} \frac{1}{4^r} \binom{2r}{r} s^{4r-3}. \quad (5.3)$$

Put  $P = -\mathcal{Q}$ , so that  $\mathcal{G} = I - \nabla P$ . Zhao's inversion formula for a Hessian-nilpotent potential gives homogeneous inversion terms [10]

$$R_m = \frac{1}{2^m m! (m+1)!} \Delta^m(P^{m+1}).$$

The degree of  $\nabla R_m$  is  $2m + 3$ . Comparing (5.3) at  $m = 2r - 3$  gives:

**Theorem 5.2.** *For every  $r \geq 2$ ,*

$$\partial_{u_1} \Delta^{2r-3}(\mathcal{Q}^{2r-2})(\bar{W}) = \frac{(2r-3)!(2r-2)!}{16} \binom{2r}{r} \neq 0.$$

*In particular,*

$$\partial_{u_1} \Delta(\mathcal{Q}^2)(\bar{W}) = \frac{3}{4}.$$

*Proof.* The exponent  $m + 1 = 2r - 2$  is even, so  $P^{m+1} = \mathcal{Q}^{m+1}$ . Multiply the coefficient of  $s^{4r-3}$  in (5.3) by  $2^m m! (m+1)!$ .  $\square$

Since  $\mathcal{Q}$  is Hessian nilpotent,  $\Delta^m(\mathcal{Q}^m) = 0$  for all  $m \geq 1$ , while theorem 5.2 gives infinitely many nonzero  $\Delta^m(\mathcal{Q}^{m+1})$ . This is an explicit failure of eventual vanishing in Zhao's framework.

## 6. A 52-TO-110 SQUARE-ZERO PAIRING INTERVAL

Let  $V = \mathbb{Q}^{19}$ , and regard

$$H \in V \otimes \text{Sym}^3(V^*).$$

Its *vector-Waring length* is the least  $R$  for which

$$H(z) = \sum_{\nu=1}^R b_\nu \ell_\nu(z)^3. \quad (6.1)$$



6.1. **The lower bound.** Let

$$W = \text{span} \{ \partial H_i / \partial z_j \} \subset \text{Sym}^2(V^*).$$

Every decomposition (6.1) gives

$$W \subset U := \text{span} \{ \ell_\nu^2 \}.$$

Exact row reduction gives

$$\dim W = 39. \tag{6.2}$$

In the coordinate order of section 3, the 12-dimensional subspace

$$\mathcal{A} = \langle z, c, q, s, k, w_1, \dots, w_7 \rangle$$

is a common zero block:

$$W|_{\mathcal{A}} = 0.$$

The coefficient system for a constant vector  $v$  satisfying  $JH(z)v = 0$  has rank 19; a specified 19-by-19 minor has determinant 18. Thus the  $\ell_\nu$  span  $V^*$ , and their restrictions span  $\mathcal{A}^*$ . Twelve independent restrictions have twelve independent squares. It follows that

$$\dim(U/W) \geq 12, \quad \dim U \geq 51.$$

**Proposition 6.1.** *Every vector-Waring decomposition of  $H$  has length at least 52.*

*Proof.* Suppose a decomposition has length at most 51. Every preceding inequality is then an equality:

$$R = \dim U = 51, \quad W = \ker(U \rightarrow \text{Sym}^2(\mathcal{A}^*)).$$

Choose a basis  $\alpha_1, \dots, \alpha_{12}$  of  $\mathcal{A}^*$  among the restricted forms. If  $\alpha = \sum c_j \alpha_j$  and  $\alpha^2$  lies in the span of the  $\alpha_j^2$ , then all  $c_i c_j$  for  $i \neq j$  vanish. Hence every restricted form is proportional to one  $\alpha_j$ .

Choose a complement  $V = \mathcal{B} \oplus \mathcal{A}$ , with  $\dim \mathcal{B} = 7$ , and let

$$\mathcal{C} = \text{pr}_{\mathcal{B}^* \otimes \mathcal{A}^*}(W).$$

The equality description forces  $\mathcal{C}$  to be invariant under the twelve coordinate projectors in the  $\alpha_j$ -basis. Therefore its right stabilizer algebra

$$\mathcal{E} = \{ T \in \text{End}(\mathcal{A}^*) : (I \otimes T)\mathcal{C} \subset \mathcal{C} \}$$

contains a conjugate of the full diagonal algebra. Its commutant can then contain no nonzero nilpotent.

Exact rational computation gives

$$\dim \mathcal{C} = 24, \quad \dim \mathcal{E} = 62, \quad \dim \mathcal{E}' = 7,$$

and  $\mathcal{E}'$  contains the matrix unit  $E_{5,0}$ , with indices numbered  $0, \dots, 11$ . This element is nonzero and square-zero, a contradiction.  $\square$

**6.2. The 110-variable upper construction.** The supplement contains rational matrices

$$B \in M_{19 \times 110}(\mathbb{Q}), \quad D \in M_{110 \times 19}(\mathbb{Q})$$

satisfying

$$\text{rank } B = \text{rank } D = 19, \quad BD = 0, \quad H(z) = B(Dz)^{*3}. \quad (6.3)$$

Put  $A = DB$  and

$$F_A(W) = W + (AW)^{*3}.$$

Then  $A^2 = 0$  and  $\text{rank } A = 19$ .

**Proposition 6.2.** *The map  $F_A: \mathbb{A}^{110} \rightarrow \mathbb{A}^{110}$  is a noninjective Keller map. It is Gorni–Zampieri paired with  $G = I + H$ .*

*Proof.* Choose a right inverse  $C$  of  $B$ . The exact data give  $AC = D$ , and

$$BF_A(Cz) = z + B(Dz)^{*3} = G(z).$$

By Sylvester’s determinant identity,

$$\begin{aligned} \det JF_A(W) &= \det \left( I_{110} + 3 \text{diag}((AW)^2)DB \right) \\ &= \det \left( I_{19} + 3B \text{diag}((DBW)^2)D \right) \\ &= \det JG(BW) = 1. \end{aligned}$$

If  $G(z_1) = G(z_2)$ , put  $u = Cz_1$ ,  $v = Cz_2$ , and  $\delta = F_A(u) - F_A(v)$ . Then  $B\delta = 0$ , hence  $A\delta = DB\delta = 0$ . Thus

$$F_A(v + \delta) = F_A(v) + \delta = F_A(u).$$

The exact collision certificate verifies  $u \neq v + \delta$ .  $\square$

**Corollary 6.3.** *For the fixed tensor  $H$ , the minimum length of a full-rank square-zero pairing satisfies*

$$52 \leq N_{\text{pair}} \leq 110.$$

*Proof.* The lower bound is proposition 6.1; every such pairing is a vector-Waring decomposition. The upper bound is proposition 6.2.  $\square$

## 7. EXACT COMPUTATION AND REPRODUCIBILITY

All example-specific calculations use exact rational arithmetic. The fresh audit reran:

- (1) the public-input-to-19D collision and nilpotency certificate;
- (2) the rank-seven factorization and Schur-complement identities;
- (3) both kernel vectors, the length-18 Jordan chain, and the degeneration certificate;
- (4) the 38D gradient identity, Hessian rank and index, inverse branch, and first Laplacian value;
- (5) the 110D matrices, square-zero identity, pairing identity, Keller determinant transfer, and collision;

- (6) the 52-cube lower-bound matrices and equality obstruction;
- (7) all 3,146 monomial tests for  $\Lambda_4$ ;
- (8) all 605 triangular monomial shifts and the ten target-cleanup directions; and
- (9) a separate verifier using only Python rational fractions for the small rank, minor, circuit, and commutant certificates.

The regenerated 724-term quartic is byte-for-byte identical to the recovered output. The 110D pairing JSON and the serialized small certificates also match their archived inputs exactly.

The source, exact serialized tensors, and successful logs are organized in `computational-supplement/`. In particular, `fixed-tensor/` contains the 19D, 38D, and 110D constructions, `continuation/` contains the Waring and compression calculations, and the two `n5-regular-*` directories separate the collision-line existence theorem from the first-normal global obstruction. The claim-to-file map and minimal replay commands are in `COMPUTATION.md`. This organization reflects the mathematical lesson of the calculation: regularity on the collision line is possible, but global nilpotence imposes new normal-direction equations.

The computations do not establish novelty, and the several SymPy scripts are not independent CAS implementations. The Fraction-only checker is independent of SymPy but consumes serialized certificate matrices. A public release should additionally reproduce the main identities in Singular, Macaulay2, Sage, or Magma.

## 8. QUESTIONS

**Question 8.1** (Minimum homogeneous dimension). *What is the minimum dimension of a cubic-homogeneous counterexample? The fixed 19-variable construction gives an upper bound, while known low-dimensional classifications and the five-dimensional restrictions collected in the companion research register leave a substantial gap.*

**Question 8.2.** *Can one improve the rank-sensitive complexity*

$$\dim K + \dim \operatorname{span} \{(K_3)_i\}$$

*by choosing a different degree-at-most-three stable representative?*

**Question 8.3** (Intrinsic tensor and pairing ranks). *What are the exact vector-Waring length, pairing rank, and square-zero realization rank of the fixed 19-variable cubic tensor? In particular, can the present vector-Waring interval between 52 and 110 be closed without committing to one reduction algorithm?*

**Question 8.4.** *Can the  $\mathfrak{sl}_{18}$  collision-monolith structure recorded in the companion research register be used to derive the Jordan types  $(18, 1)$  and  $(35, 2, 1)$  without coefficient calculation?*

## APPENDIX A. ONE COVER IN SEVERAL NORMAL FORMS

The reductions below do more than transport a collision: after adjoining purely transcendental variables, they preserve the function-field extension induced by the original map. Thus the cubic, symmetric, and square-zero descendants are presentations of the same generically three-sheeted cover.

**Proposition A.1** (Rees presentation of the rank-sensitive suspension). *Let  $K = X + Q + C$ , let  $C = Bq$ , and let*

$$G(X, w, t) = (X + tQ(X) + t^2Bw, w - q(X), t).$$

*Define triangular automorphisms*

$$S(X, v, t) = (X, v + q(X), t), \quad T(Y, v, t) = (Y - t^2Bv, v, t).$$

*Then*

$$T \circ G \circ S = (K_t \times \text{id})(X, v, t), \quad K_t(X) = X + tQ(X) + t^2C(X) = t^{-1}K(tX).$$

*Consequently, after inverting  $t$ , the function-field extension induced by  $G$  is obtained from that induced by  $K$  by adjoining independent transcendental variables. In particular, generic degree and geometric monodromy are preserved.*

*Proof.* Direct substitution gives

$$G(X, v + q(X), t) = (X + tQ(X) + t^2Bv + t^2C(X), v, t),$$

and the target shear  $T$  removes  $t^2Bv$ . Over  $k(t)$  the change of variables  $U = tX$  identifies  $K_t$  with  $K$ .  $\square$

**Proposition A.2** (The symmetric double is a stable presentation). *Let  $F = X + H$  be Keller and put*

$$\Phi_F(x, y) = (F(x), JF(x)^T y).$$

*Then the polynomial automorphism*

$$S_F(x, y) = (x, JF(x)^T y)$$

*satisfies*

$$\Phi_F = (F \times \text{id}) \circ S_F.$$

*Hence the symmetric double, and therefore its linear gradient twist, induces the same function-field extension as  $F$  after adjoining  $\dim F$  independent variables.*

*Proof.* Since  $\det JF = 1$ , the inverse matrix  $JF^{-T}$  has polynomial entries, so  $S_F$  is a polynomial automorphism.  $\square$

**Proposition A.3** (A full-rank square-zero pairing is stably right-equivalent). *Suppose*

$$H(z) = B(Dz)^{*3}, \quad BD = 0,$$

where  $B : W \rightarrow V$  is surjective and  $D : V \rightarrow W$  is injective. Put  $A = DB$  and  $F_A(w) = w + (Aw)^{*3}$ . Choose a right inverse  $C$  of  $B$  and write  $W = C(V) \oplus E$ , where  $E = \ker B$ . Set

$$\rho(z) = (Dz)^{*3} - CH(z) \in E.$$

In the coordinates  $w = Cz + \eta$ , one has

$$F_A(z, \eta) = (G(z), \eta + \rho(z)), \quad G = I + H.$$

Precomposition by  $(z, \eta) \mapsto (z, \eta - \rho(z))$  therefore changes  $F_A$  into  $G \times \text{id}_E$ .

**A.1. The quotient cubic and its monodromy.** For the original three-variable map, write its target coordinates as  $(\xi, \eta, \zeta)$ . The source has the grading  $\text{wt}(x, y, z) = (1, -1, -2)$ , and the target weights are  $(-2, -1, 1)$ . Put

$$\nu = xy, \quad v = x^2z, \quad r = 1 + \nu, \quad c = 2 - 3\nu - v, \quad q = cr,$$

and form the target invariants

$$A = \xi\zeta^2, \quad B = \eta\zeta.$$

A direct calculation gives

$$A = q^2 + cq - q^3, \quad B = 4q + 2c - 3q^2.$$

Eliminating  $c$  yields the cubic

$$q^3 - 2q^2 + Bq - 2A = 0. \tag{A.1}$$

Its discriminant is

$$-4(27A^2 - 18AB + 16A + B^3 - B^2) = -4\zeta^2 \mathcal{N}(\xi, \eta, \zeta),$$

where

$$\mathcal{N}(\xi, \eta, \zeta) = 27\xi^2\zeta^2 - 18\xi\eta\zeta + 16\xi + \eta^3\zeta - \eta^2. \tag{A.2}$$

The quadratic polynomial in  $A$  inside the discriminant has discriminant  $-4(3B - 4)^3$ , hence is not a square over  $\mathbb{Q}(B)$ . Together with the generic irreducibility of (A.1), this gives geometric monodromy  $S_3$ .

## APPENDIX B. A TWO-PARAMETER FAMILY OF ALL-ORDER INVERSE RAYS

Let  $e_x, e_y, e_{w_1}, e_t$  refer to the indicated coordinates of the 19-variable cubic map, and for  $\alpha, \beta \in \mathbb{Q}$  put

$$Y_{\alpha, \beta} = \alpha e_x + e_y + \beta e_{w_1} + e_t.$$

Let  $\widehat{G}_0^{-1}$  denote the formal inverse branch at the origin.

**Theorem B.1** (Exact inverse-ray family). *For all  $\alpha, \beta$ ,*

$$(\widehat{G}_0^{-1}(sY_{\alpha, \beta}))_x = \frac{(1 - 2\alpha s^4 + 2\beta s^6)^{-1/2} - 1}{s^3},$$

where the square root is the branch with constant term 1.

*Proof.* The  $t$ -coordinate gives  $t = s$ , and the  $w$ -coordinates give  $w = q(X) + s\beta e_1$ . Therefore

$$K(sX) = s^2(\alpha e_x + e_y) - s^4\beta e_x.$$

After applying the linear normalization and eliminating the stable auxiliary coordinates, this is the original three-variable target curve

$$(\xi, \eta, \zeta) = (0, s^2, 2\alpha s^2 - 2\beta s^4).$$

On the local branch above  $\xi = 0$ , write

$$R = (1 + xy)^2 z + y^2(4 + 3xy).$$

The equation  $\xi = (1 + xy)R = 0$  gives  $R = 0$ , and then  $\eta = y = s^2$ . With  $r = 1 + xy$ , the third component simplifies to

$$\zeta = \frac{1 - r^{-2}}{y}.$$

Thus

$$r = (1 - y\zeta)^{-1/2} = (1 - 2\alpha s^4 + 2\beta s^6)^{-1/2},$$

and the original  $x$ -coordinate is  $(r - 1)/s^2$ . Since the original source is  $sX$ , division by  $s$  gives the formula.  $\square$

The same calculation explains the singularities: along this curve,

$$\mathcal{N}(0, s^2, 2\alpha s^2 - 2\beta s^4) = -s^4(1 - 2\alpha s^4 + 2\beta s^6).$$

Thus the algebraic singularities of the inverse ray are precisely its intersections with the discriminant/nonproperness hypersurface.

Let  $\mathcal{G} = I + \nabla \mathcal{Q}$  be the 38-variable gradient descendant, and put

$$W_{\alpha, \beta} = T^{-1}(Y_{\alpha, \beta}, 0).$$

For  $n \geq 0$  define

$$c_n(\alpha, \beta) = [u^n](1 - 2\alpha u^2 - 2\beta u^3)^{-1/2}.$$

For  $n \geq 2$  one has the finite sum

$$c_n(\alpha, \beta) = \sum_{k=\lceil n/3 \rceil}^{\lfloor n/2 \rfloor} 2^{-k} \binom{2k}{k} \binom{k}{n-2k} \alpha^{3k-n} \beta^{n-2k}. \quad (\text{B.1})$$

In particular,  $c_n(\alpha, \beta) > 0$  for positive rational  $\alpha, \beta$ .

**Theorem B.2** (Nonvanishing at every Laplacian order). *For every  $m \geq 1$  and every positive rational  $\alpha, \beta$ ,*

$$\partial_{u_1} \Delta^m(\mathcal{Q}^{m+1})(W_{\alpha, \beta}) = 2^{m-1} m!(m+1)! c_{m+3}(\alpha, \beta) > 0.$$

*Thus the same quartic violates the eventual-vanishing conclusion at every order, not merely along an infinite parity class.*

*Proof.* The first coordinate of the inverse gradient ray is half the first coordinate in Theorem B.1. If

$$(1 - 2\alpha u^2 + 2\beta u^3)^{-1/2} = \sum d_n u^n,$$

then  $d_n = (-1)^n c_n(\alpha, \beta)$ . Compare the coefficient of  $s^{2m+3}$  with Zhao's inversion formula for  $P = -Q$ :

$$R_m = \frac{1}{2^m m! (m+1)!} \Delta^m (P^{m+1}).$$

The two signs cancel, giving the displayed positive formula.  $\square$

The central-binomial formula in the earlier draft is the boundary case  $(\alpha, \beta) = (1/2, 0)$ . Turning on  $\beta$  fills the parity gaps.

### APPENDIX C. THE NILPOTENT FLAG MOVES

Let  $A(Z) = JH(Z)$  for the 19-variable cubic tensor.

**Proposition C.1** (A two-point obstruction to strong nilpotence). *Although  $A(Z)$  is nilpotent for every  $Z$ , the family is not strongly nilpotent and is not simultaneously strictly triangularizable. Indeed,*

$$A(e_d)A(e_t)|_{\langle e_{w_3}, e_{w_6} \rangle} = \begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix},$$

and all other entries of the product vanish. Hence

$$\chi_{A(e_d)A(e_t)}(\lambda) = \lambda^{18}(\lambda + 3).$$

This gives a concrete form of the distinction between pointwise nilpotence and a common nilpotent flag: the generic matrix is almost regular nilpotent, but its Jordan flag twists with the base point.

#### C.1. A self-dual extension lemma.

**Lemma C.2** (Isotropic terminal vector). *Let  $A$  have Jordan type  $(m, 1)$ , let  $\ker A = \langle v_1, v_2 \rangle$ , and suppose  $\operatorname{im} A^{m-1} = \langle v_2 \rangle$ . Let  $L = L^T$  and*

$$N = \begin{pmatrix} A & 0 \\ L & A^T \end{pmatrix}.$$

Define the symmetric form

$$\beta(u, v) = u^T L v \quad (u, v \in \ker A).$$

If  $\beta$  has rank one with radical  $\langle v_2 \rangle$ , if  $N^{2m-2} \neq 0$ , and if  $\beta(v_2, v_2) = 0$ , then  $N$  has Jordan type  $(2m-1, 2, 1)$ .

*Proof.* Projection of  $\ker N$  to its first factor identifies its image with  $\operatorname{rad} \beta$ , and each fiber is  $\ker A^T$ . Thus  $\dim \ker N = 2 + 1 = 3$ . The lower-left block of  $N^{2m-1}$  is the sole possible term

$$(A^T)^{m-1} L A^{m-1},$$

which vanishes because the terminal line is isotropic. The preceding power does not vanish:

$$N^{2m-2} \neq 0.$$

Thus the nilpotency index is  $2m - 1$ . The operator acts on a space of dimension  $2m + 2$  and has three blocks. Its largest block has size  $2m - 1$ , so its partition is  $(2m - 1, 2, 1)$ .  $\square$

For the present tensor, exact differentiation gives

$$D^2H[v_1, v_1] = -4t^3xz e_x, \quad D^2H[v_1, v_2] = D^2H[v_2, v_2] = 0.$$

Hence the form induced by  $L = \text{Hess}_x(y^T H(x))$  has rank one on  $\ker A$ , with radical  $\langle v_2 \rangle$ . The previously verified nonzero entry of  $N^{34}$  then applies Lemma C.2 with  $m = 18$  and explains the partition  $(35, 2, 1)$  as a one-box degeneration of maximal chain gluing.

#### APPENDIX D. A ROBUST EQUIVARIANT COMPRESSION OBSTRUCTION

The normalized 11-variable map is equivariant for the weights

$$(1, -1, -2, 0, 1, -2, 0, -1, -2, 2, -1)$$

on  $(x, y, z, a, b, c, d, q, s, h, k)$ . Let  $\mathfrak{X}_2^{(0)}$  be the space of weight-preserving quadratic vector fields; exact counting gives  $\dim \mathfrak{X}_2^{(0)} = 115$ .

For a quadratic formal coordinate change  $I + P$ , the cubic jet is  $C + [Q, P]$ . Its quartic term is

$$\mathcal{O}_4(P) = [C, P] + \frac{1}{2}D^2Q[P, P] - JP[Q, P] - \frac{1}{2}D^2P[Q, Q]. \quad (\text{D.1})$$

Put

$$\Sigma_{-2} = \langle y^2, qy, ky, az, dz, ac, as, cd, ds, q^2, kq, k^2 \rangle$$

and

$$P_\sigma = -d^2e_a + \sigma e_s \quad (\sigma \in \Sigma_{-2}).$$

**Proposition D.1** (A positive-dimensional obstructed compression family). *For every  $\sigma \in \Sigma_{-2}$ , the cubic jet  $C + [Q, P_\sigma]$  is supported only in the six output coordinates  $x, y, z, b, c, s$ , and therefore has coordinate-span rank at most six. Nevertheless,*

$$\Lambda_4(\mathcal{O}_4(P_\sigma)) = 1.$$

*Hence no cubic source correction removes the quartic term anywhere in this 12-parameter formal family. Excluding the two monomials involving  $s$  gives a 10-parameter family of actual triangular polynomial automorphisms.*

A larger exact calculation gives the following natural slice. Let

$$\mathcal{S} = \{P \in \mathfrak{X}_2^{(0)} : (C + [Q, P])_a = (C + [Q, P])_d = (C + [Q, P])_q = 0, \\ (C + [Q, P])_h = (C + [Q, P])_k = 0\}.$$

Then  $\mathcal{S}$  is a nonempty affine space of dimension 20, and

$$\Lambda_4(\mathcal{O}_4(P)) = 1 \quad (P \in \mathcal{S}).$$



The tangent space at  $P_0 = -d^2 e_a$  to the equivariant rank-at-most-six locus has dimension 22; the two extra directions are precisely the infinitesimal weight-preserving target changes that add the surviving  $y$ -row to the  $q$ - and  $k$ -rows. Thus  $\mathcal{S}$  is the expected 20-dimensional transverse slice modulo linear target reparametrization.

This changes the interpretation of the obstruction: cubic compression is not isolated or difficult. It occurs in a family, while a secondary quartic class is constant and nonzero across the family.

#### APPENDIX E. FULL-KERNEL PENCILS AND THE $19 \rightarrow 18$ COMPRESSION PROBLEM

Two computations continue our dimension-five and nineteen-variable analyses. The first reorganizes the regular full-kernel pencil by binary-sextic invariant theory. The second proves a no-lift theorem on a natural transversal to the  $19 \rightarrow 18$  compression locus.

**E.1. The regular full-kernel chart.** In the full-kernel chart write

$$M(s, t) = R(s, t)T^{-1}K(s, t).$$

With the standard quartic Veronese column  $k$ , cubic column  $q$ , and row  $\ell$ , exact multiplication gives

$$N_4^3 = 6q\ell, \quad Rq = a_7 k,$$

and hence

$$M(s, t)^4 = 6a_7 k(s, t)\ell(s, t)T^{-1}K(s, t).$$

**Proposition E.1** (Exact regularity criterion). *On  $\det T \neq 0$ , the pencil  $M(s, t)$  has Jordan type (5) at every point of  $\mathbb{P}^1$  if and only if  $a_7 \neq 0$ .*

*Proof.* All factors in the displayed formula other than  $a_7$  are nonzero at every point of  $\mathbb{P}^1$ , while  $M^5 = 0$ . Thus  $M^4$  is everywhere nonzero exactly when  $a_7 \neq 0$ .  $\square$

Put  $\lambda = a_7$ . After a fixed linear change of the other seven parameters,

$$T = \frac{\lambda}{4}I + S_b, \quad S_b^T J + JS_b = 0,$$

and the  $b$ -space is the irreducible principal- $\mathfrak{sl}_2$  module

$$\mathrm{Sym}^6(\mathbb{C}^2).$$

Thus the chart is naturally

$$\mathbb{C}\lambda \oplus \mathrm{Sym}^6(\mathbb{C}^2),$$

not an unstructured eight-parameter matrix family. If  $A, B$  are the standard degree-two and degree-four binary-sextic invariants in the chosen normalization, then

$$\det T = \frac{(\lambda^2 + 4A)^2 - 96B}{256}.$$

The marking equations are linear in the nine unfixed marking variables. On an explicit pivot chart they eliminate to one degree-12 parameter equation and five degree-13 equations. A candidate invariant formula for the degree-12 equation was reconstructed by exact modular interpolation and checked on rational samples; it is retained in the supplement as a computational conjecture until a coefficient certificate or conceptual invariant-theoretic derivation is supplied.

**Proposition E.2** (Finite-field sample audit). *An exhaustive  $\mathbf{F}_{13}$ -calculation finds thirteen rational points of this normalized collision system. Together with the previously checked six points over  $\mathbf{F}_7$  and eleven over  $\mathbf{F}_{11}$ , all thirty sampled points have nonzero first-normal obstruction. This is exact finite-field evidence, not a proof that the obstruction section is nowhere zero on the characteristic-zero collision curve.*

*Exact computation.* The finite-field programs enumerate the normalized solutions and evaluate the first-normal obstruction in the corresponding prime field. Their saved exact outputs give the three counts and a nonzero value at every listed point.  $\square$

**E.2. The unrestricted compression tangent space.** Let  $K = X + Q + C: \mathbb{A}^{11} \rightarrow \mathbb{A}^{11}$  be the normalized degree-at-most-three map used in the fixed nineteen-variable construction. A quadratic source field  $P$  changes its cubic term to

$$C_P = C + [Q, P].$$

**Proposition E.3** (Large row-killing families). *The affine space of quadratic source fields has dimension 726. There is a 109-dimensional affine family on which five prescribed cubic output rows vanish and the remaining coordinate span has rank at most six. The quartic obstruction functional is identically one on this family. Inside it, a specified 75-dimensional family consists of triangular polynomial automorphisms. The full rank-six tangent space at the base point has dimension 129, leaving a 20-dimensional quotient by the row-killing directions; a nonempty Zariski-open subset of that quotient is obstructed at second order.*

*Exact computer-assisted proof.* The assertions are rational ranks and symbolic bracket identities in the displayed coefficient bases. Fixed full-rank minors certify the ranks, and the triangular family is obtained by removing the 145 forbidden output coefficients from the 220-dimensional triangular ansatz.  $\square$

**E.3. The complete transverse cone.** At the row-killing rank-six point

$$P_* = 2hx e_z - d^2 e_a - 6ax e_b + (2az - \frac{2}{3}bh)e_c + 2b^2 e_s - h^2 e_k,$$

the determinantal tangent space has dimension 115. Modulo the 109 row-killing directions, choose transverse coordinates  $u_0, \dots, u_5$ .

**Theorem E.4** (Transverse no-lift). *In this six-dimensional transversal, the second-order rank-six cone is the reduced scheme*

$$V(u_0u_5, u_2u_3, u_3u_5) = V(u_3, u_5) \cup V(u_0, u_3) \cup V(u_2, u_5).$$

*There is an explicit polynomial family  $P(u)$  such that*

$$\text{rank}(C + [Q, P(u)]) \leq 6 \iff u_0u_5 = u_2u_3 = u_3u_5 = 0.$$

*Every point of the cone is therefore integrated to an exact cubic jet of coordinate span six. Nevertheless no member lifts through quartic order.*

*Exact computer-assisted proof.* The projected second fundamental form has the three displayed generators. Five exact pivot charts show that the Schur-complement ideal of the explicit family is exactly their radical intersection and that six output rows retain rank six everywhere on it.

For a quartic vector field  $E$ , set

$$\Gamma_4(E) = [h^4](E_z).$$

Support alone gives  $\Gamma_4([Q, P_3]) = 0$  for every cubic source correction  $P_3$ . Direct exact calculation on the whole six-parameter family gives

$$\Gamma_4(\mathcal{O}_4(P(u))) = -1.$$

Thus the quartic homological equation cannot hold at any rank-six member.  $\square$

The theorem resolves the complete natural six-dimensional transversal, but not the entire 726-parameter compression problem: nonlinear coupling with the 109 row-killing parameters, independent target changes, and stable-coordinate changes remain outside its scope. A previously studied two-parameter active surface is superseded by theorem E.4 and is not used.

#### COMPANION RESULTS AND RESEARCH REGISTER

This reader edition is accompanied by a separate *Results and Research Register*. The register preserves secondary proved results, conditional developments, open questions, corrections, and computational leads without making them part of this paper's theorem spine. Proof-bearing technical developments omitted from this reader PDF remain in the versioned source tree and in the complete version-8 archival edition. The v9 primary-home map distinguishes the paper responsible for a result from other papers that use or motivate it.

#### AI AND COMPUTATIONAL DISCLOSURE

AI systems were used extensively in exploration, symbolic-program development, proof drafting, source organization, and manuscript editing. They are not authors. The public 11-variable source also discloses that its explicit construction and verification code were generated by ChatGPT [2]. The

evidence for the computer-assisted theorems is the exact certificate suite described in section 7, not language model output. Human specialist review has not yet been completed.

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